Deep Learning-Based Deepfake Detection

M.Tech Sem 2 | DMML
Minor Project

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Problem Statement

To design and implement a deep learning-based system
that detects whether a given video is real or deepfake
using spatial and temporal features.

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02 System Architecture
03 Dataset Exploration
04 Pre-processing
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INTRODUCTION

What’s Deepfake

LINK

Deepfakes are synthetic media generated using deep learning techniques where a person in an existing image or video is replaced with someone else's likeness.

They are commonly created using Generative Adversarial Networks (GANs),
which consist of a generator and a discriminator network.

Real or Fake?

Deepfakes often contain visual artifacts such as unnatural blinking, inconsistent lighting, or facial distortions.

Real
Fake

Fake
Fake

Fake
Real

[Rössler et al. 2019]

Source

Why Deep Fake Detection ?

- Fake News
- Financial fraud
- Misuse of celebrity identities
- Politician videos
- Cybersecurity threats

How Deep Fakes Are Created ?

Deepfakes are generated using encoder-decoder architectures or GAN-based models that learn facial representations and reconstruct them onto target videos.

Tools for deep fake creation.

- Faceswap
- Faceit
- DeepFaceLab
- DeepfakeCapsuleGAN
- Large resolution facemasked

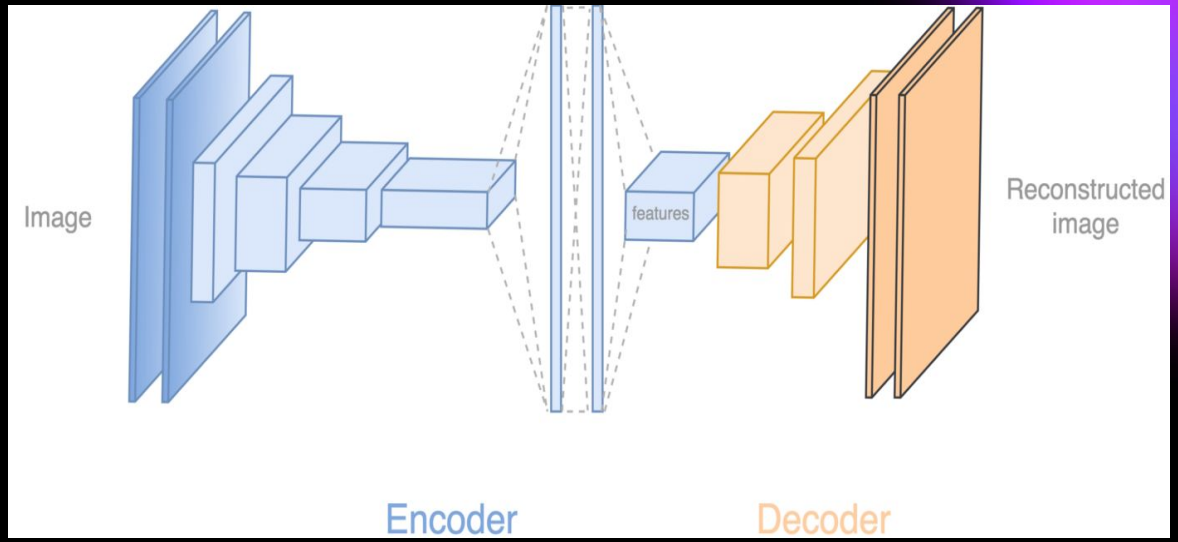


Figure 1 : Deepfake generation

Literature Review

Paper Title	Authors	Technique Used	Key Idea	Limitations
Exposing DeepFake Videos by Detecting Face Warping Artifacts	Yuezun Li, Siwei Lyu	CNN	Detects visual inconsistencies between face region and surrounding area caused by warping	Does not consider temporal information
Exposing AI Created Fake Videos by Detecting Eye Blinking	Yuezun Li, Ming-Ching Chang, Siwei Lyu	LRCN (CNN + RNN)	Uses abnormal eye blinking patterns to identify deepfakes	Modern deepfakes can simulate blinking
Deepfake Video Detection Using Recurrent Neural Networks	Darius Güera, Edward J. Delp	CNN + RNN	Combines frame-level features with temporal sequence modeling	Uses small dataset, limited generalization

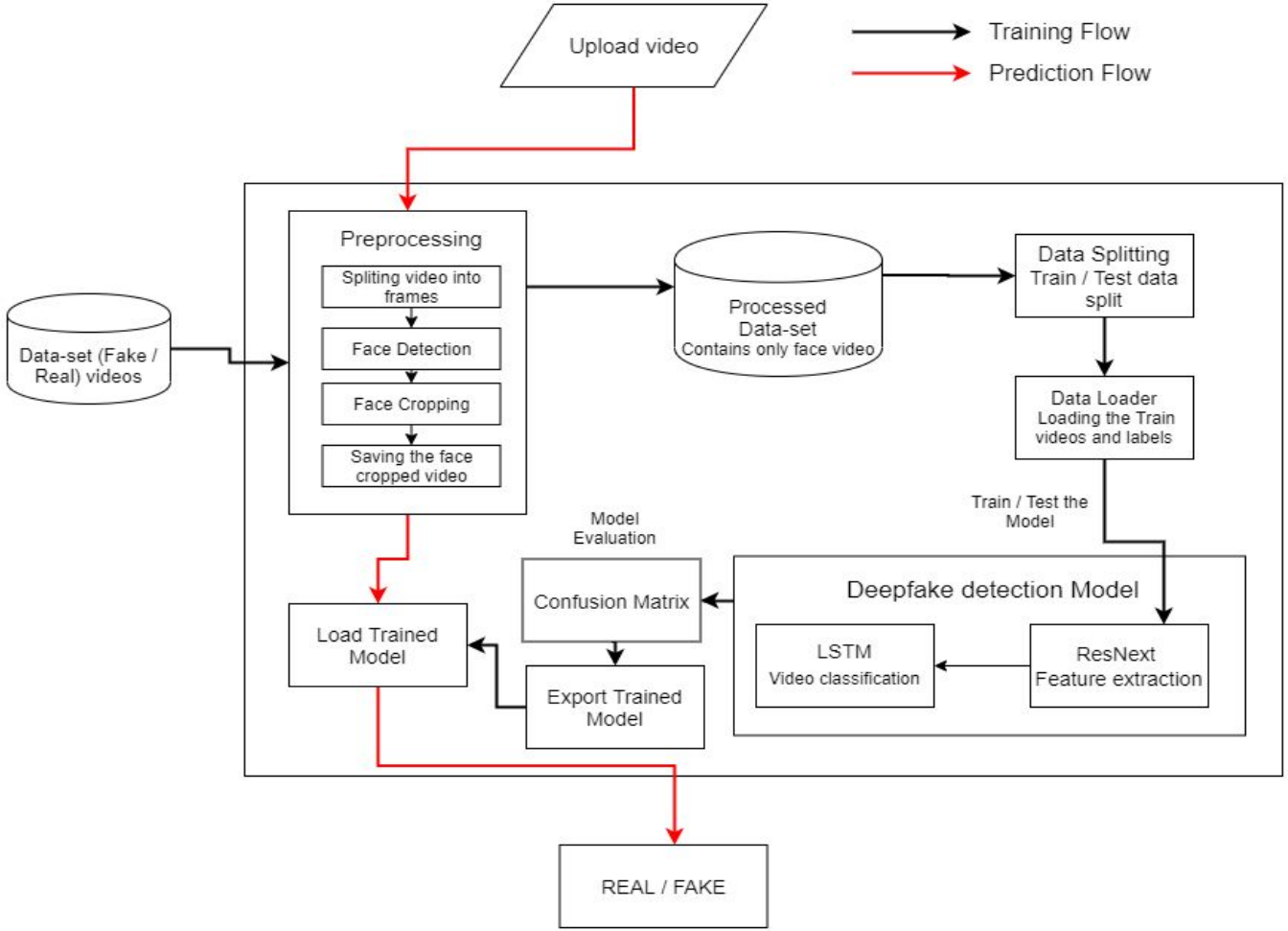


Figure 2 : System Architecture

SYSTEM ARCHITECTURE

The system processes input videos by extracting frames, detecting faces, and generating cropped face sequences.
These sequences are passed through a CNN for feature extraction
and an LSTM for temporal classification.

DATA-SET EXPLORATION

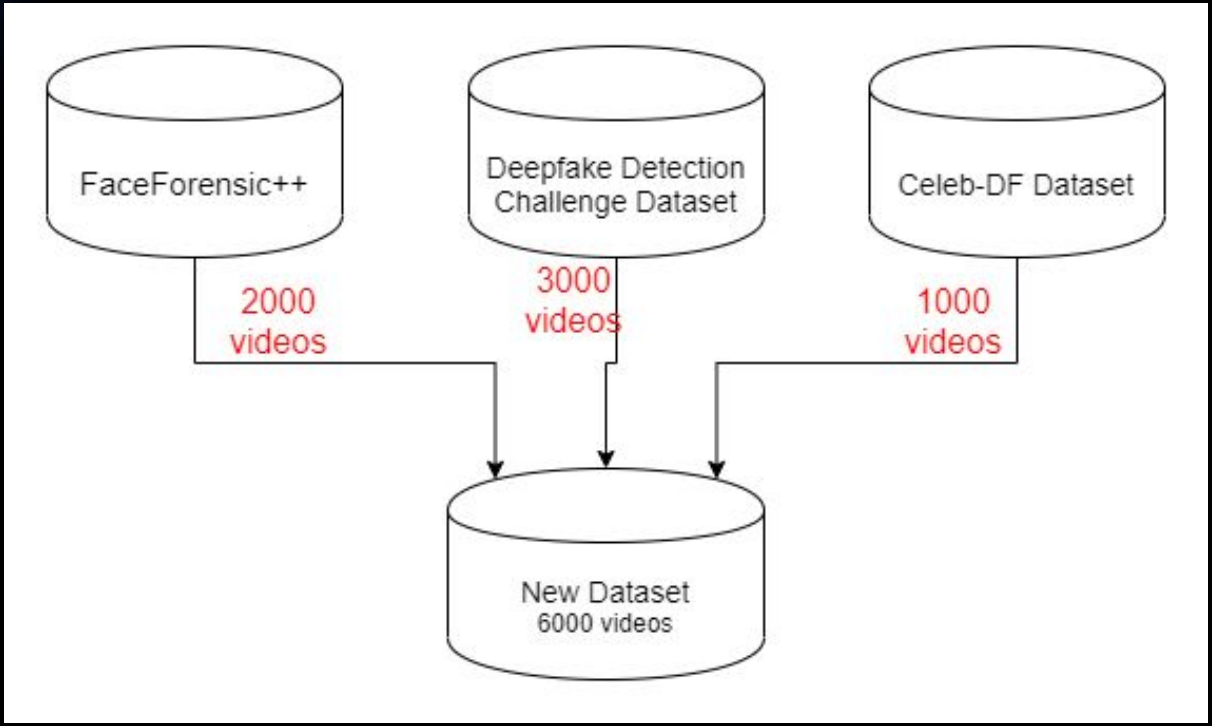


Figure 3 : Dataset Exploration

Total dataset: 6000 videos
Train/Test split: 70/30
Balanced dataset (real vs fake)

PRE-PROCESSING

1. SPLIT VIDEO INTO FRAMES
2. FACE DETECTION
3. CROPPING FACE
4. CREATING NEW FACE CROPPED VIDEO (112x112)
5. SAVING THE FACE CROPPED VIDEO

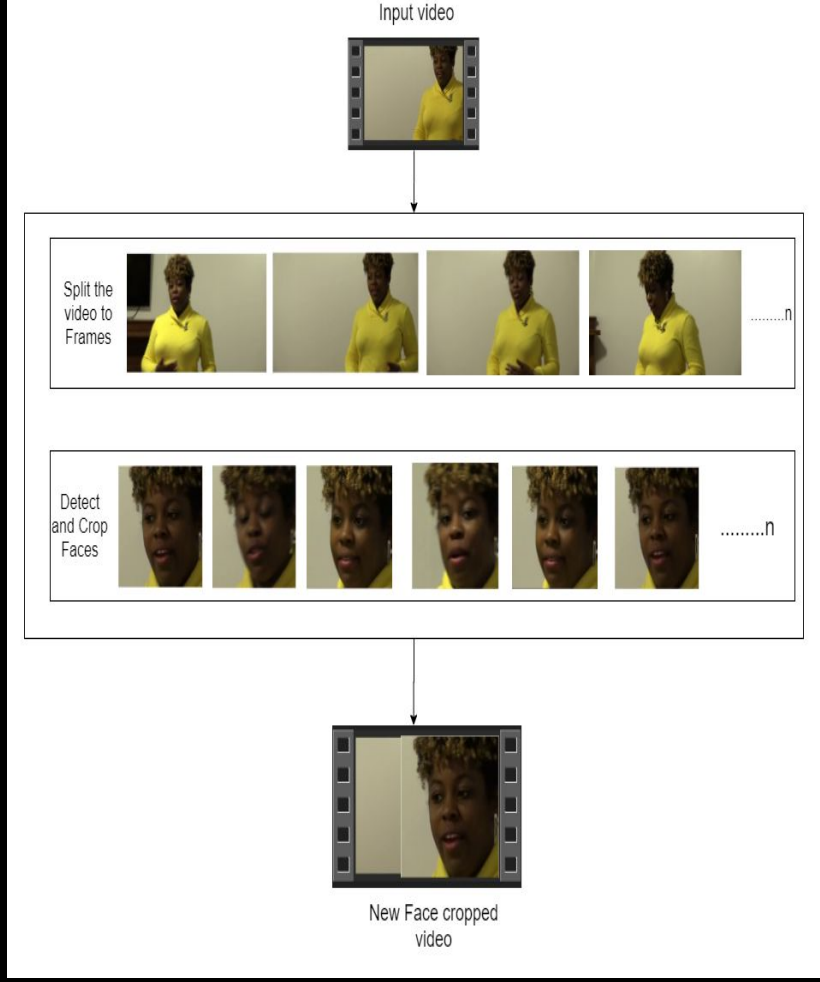


Figure 4 : Pre-processing of video

Model Architecture

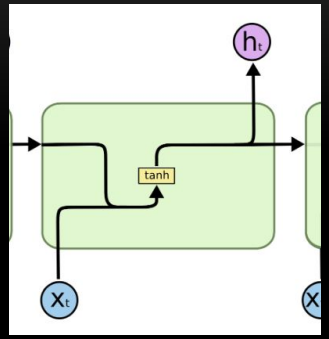
Fully Connected Layer → Classification (Real / Fake)

CNN: ResNeXt-50 (Feature Extraction)→ Extracts spatial features from each frame

stage	output	ResNeXt-50 (32×4d)
conv1	112×112	7×7, 64, stride 2
conv2	56×56	3×3 max pool, stride 2
		[1×1, 128 ; 3×3, 128, C=32 ; 1×1, 256] ×3
conv3	28×28	[1×1, 256 ; 3×3, 256, C=32 ; 1×1, 512] ×4
conv4	14×14	[1×1, 512 ; 3×3, 512, C=32 ; 1×1, 1024] ×6
conv5	7×7	[1×1, 1024 ; 3×3, 1024, C=32 ; 1×1, 2048] ×3
	1×1	global average pool ; 1000-d fc, softmax
# params.		**25.0**×10^6

Sequential Layer

LSTM:
→ Captures temporal dependencies across frames



Activation: ReLU
Dropout: 0.4
Loss: Binary Cross Entropy

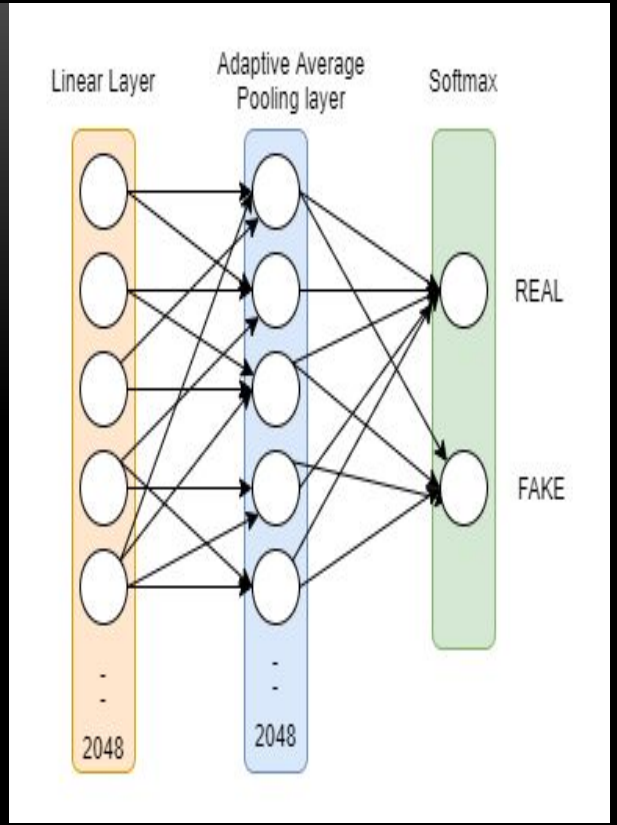


Figure 5 : Model Architecture

TRAINING WORKFLOW

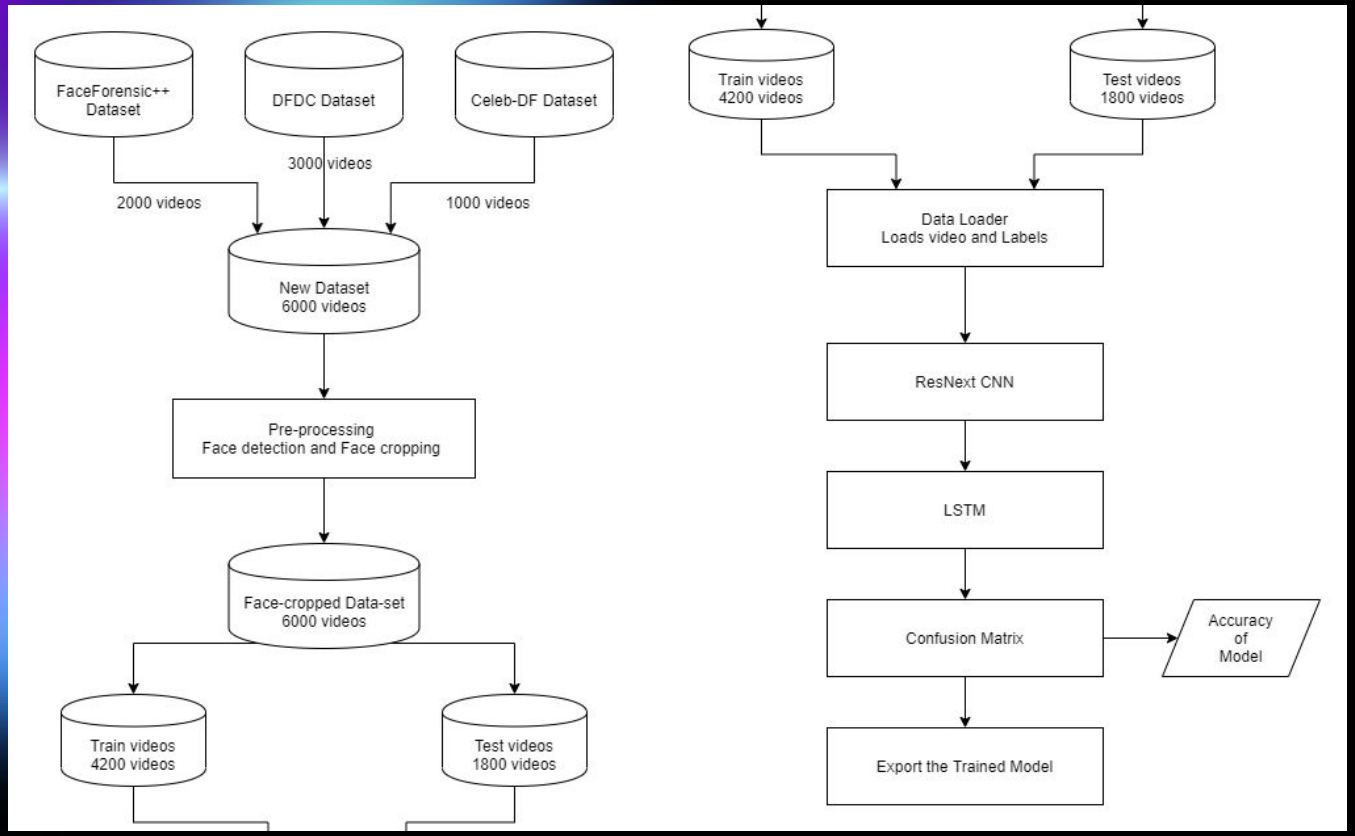
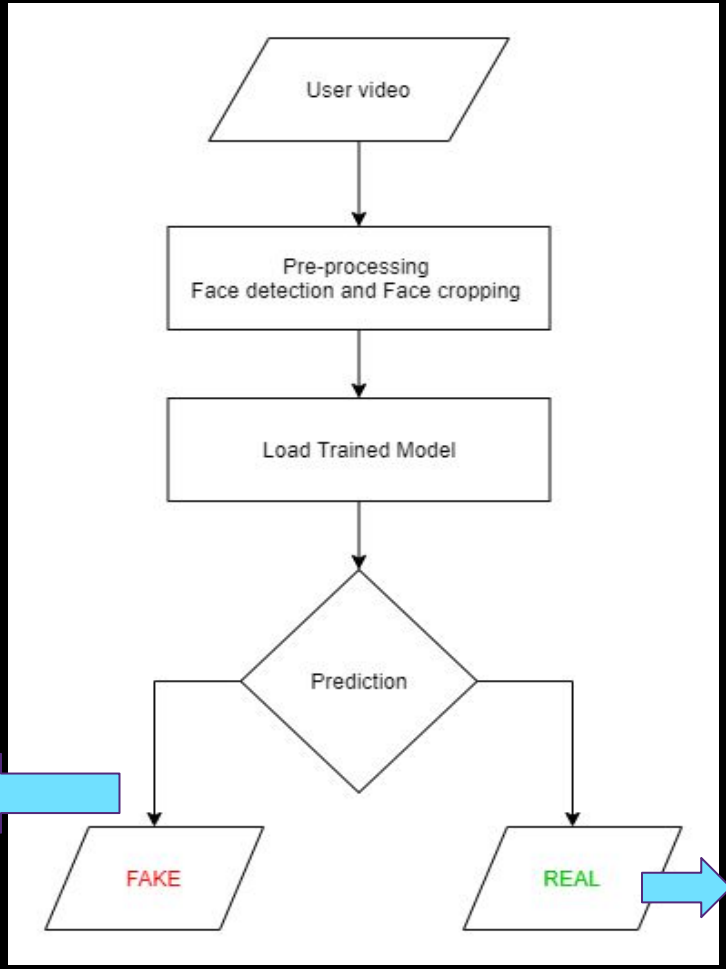


Figure 6 : Training Flow [1]

- **Epochs**: 15
- **Batch Size**: 2
- **Optimizer**: **Adam** (with a weight decay of 1e-5)
- **Learning Rate**: **1e-5** (0.00001)

PREDICTION WORKFLOW



Output: Softmax probabilities
(Fake: 0.xx, Real: 0.xx)

Prediction based on highest probability

Figure 7 : Prediction Flow [1]

RESULTS

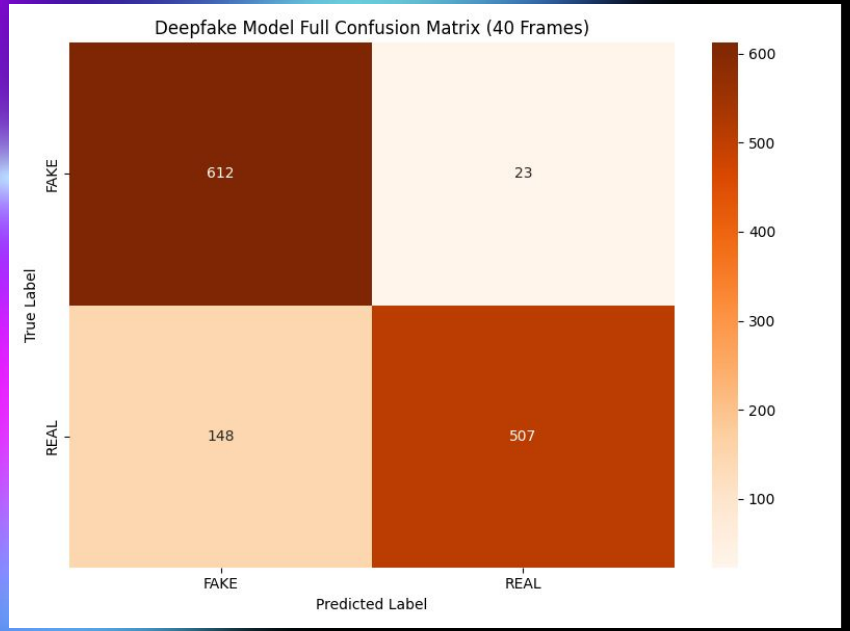


Figure 8 : Confusion Matrix (40 Frames)

Evaluated Model Profile	Frame Sequence Length	Overall Accuracy(%)	Precision (%)	Recall (%)	F1-Score (%)
model_v2_acc_10_frames	10 Frames	83.18	83.02	83.35	84.87
model_v2_acc_20_frames	20 Frames	84.04	83.87	84.21	85.89
model_v2_acc_40_frames	40 Frames	86.74	86.66	86.89	88.09
model_v2_acc_60_frames	60 Frames	86.28	86.17	86.44	87.85
model_v2_acc_80_frames	80 Frames	86.59	86.49	86.74	88.07
model_v2_acc_100_frames	100 Frames	86.74	86.64	86.90	88.28

Table 1 : Results

LIVE DEMO

CONCLUSION & FUTURE WORK

Conclusion:
- Developed deepfake detection system using CNN + LSTM
- Achieved ~86.74% accuracy
- Temporal modeling improves performance
- 40-frame sequence gives best results

Future Work:
- Use Vision Transformers
- Improve real-time performance
- Extend to audio deepfake detection

REFERENCES

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THANK YOU
Do you have Any Question?